Making Health Data ACTIONABLE

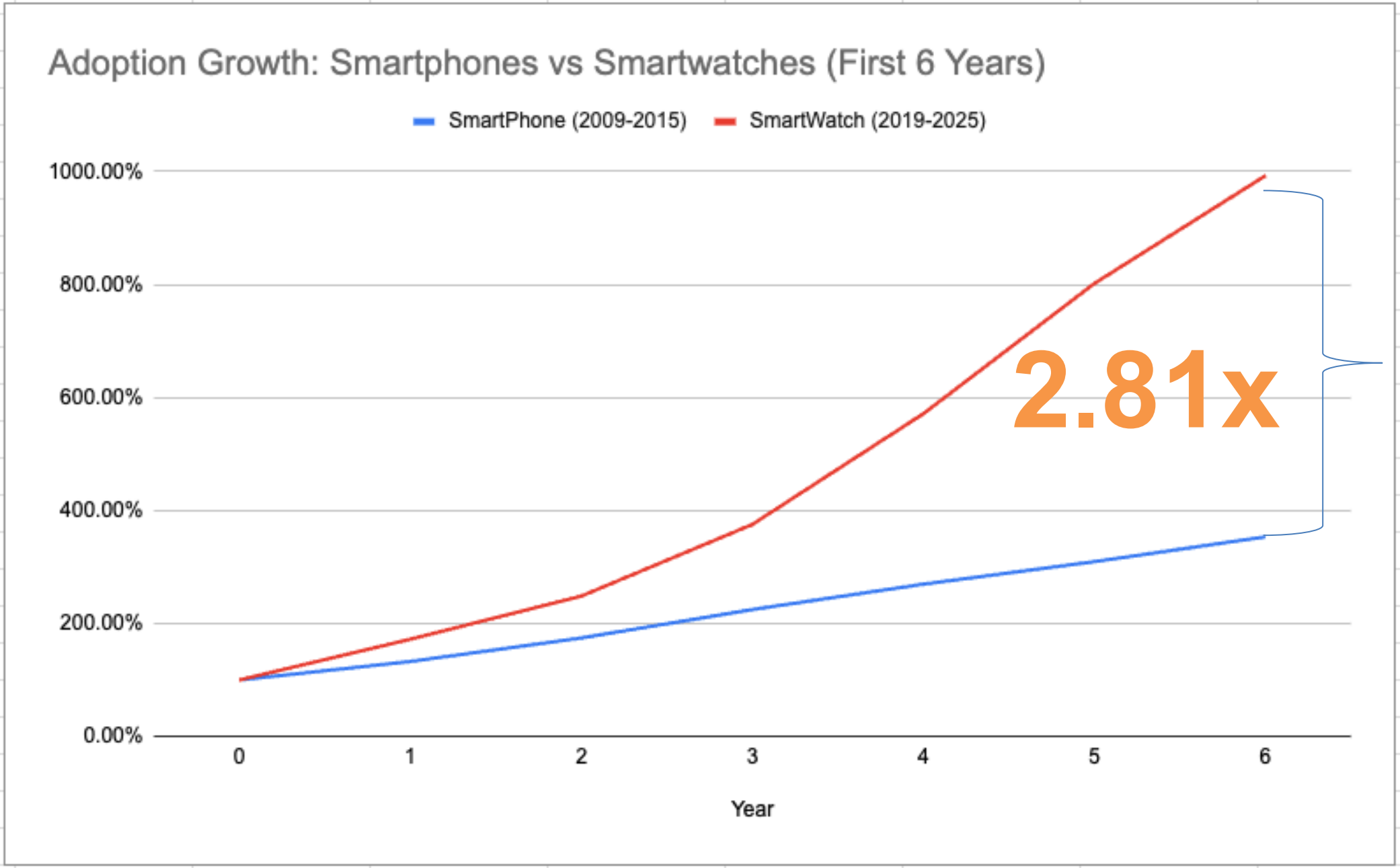
Group 5
Haoming Liu, Zhangxizi Qiu
Mridul Kumar Sharama,
Wing Chun Leung

10:41
2.14 mi
19:38
186 Peak bpm
09'36" current

?

9.92x

Growth Rate of SmartWatch Users (2019 - 2025)


Adoption Growth: Smartphones vs Smartwatches (First 6 Years)
SmartPhone (2009-2015)
SmartWatch (2019-2025)
1000.00%
800.00%
600.00%
400.00%
200.00%
0.00%
0
1
2
3
4
5
6
Year
2.81x

Dynamic patterns of healthy lifestyle awareness after COVID-19: a study using Google Trends and joinpoint regression

Zahroh Shaluhiyah [1,*], Shabrina Arifia Qatrannada
Sri Handayani [4], Dzul Fahmi Afrivanto [1], Usha Rani

Author infor

PMCID: PMC12

The Impacts of COVID-19 on Physical Activity, Dietary Changes and Health Awareness of University Students in Hong Kong: A Cross-Sectional Study

Who we

ng[1], Sigmund Luey[1],
d[1]

British Heart Foundation

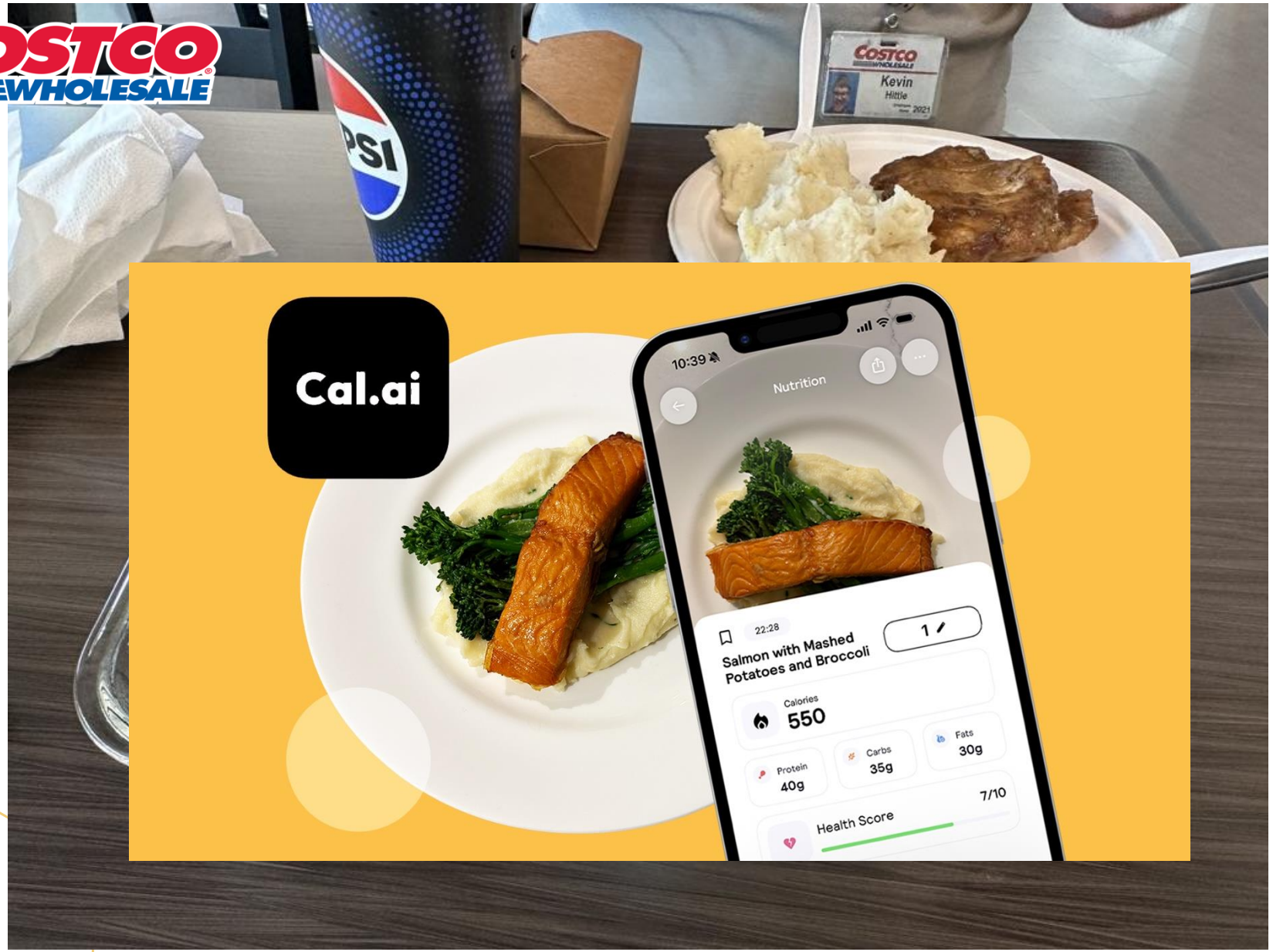
Donate

What we do > News from the BHF > News archive > Boom expected in post pandemic fitness

FUNDRAISING

Boom expected in post pandemic fitness according to our new survey

by Emily Hickey-Mason Published: 14 May 2021


Cal.ai
10:39
Nutrition
22:28
Salmon with Mashed Potatoes and Broccoli
1
Calories
550
Protein
40g
Carbs
35g
Fats
30g
Health Score
7/10

...SO?

FAKE

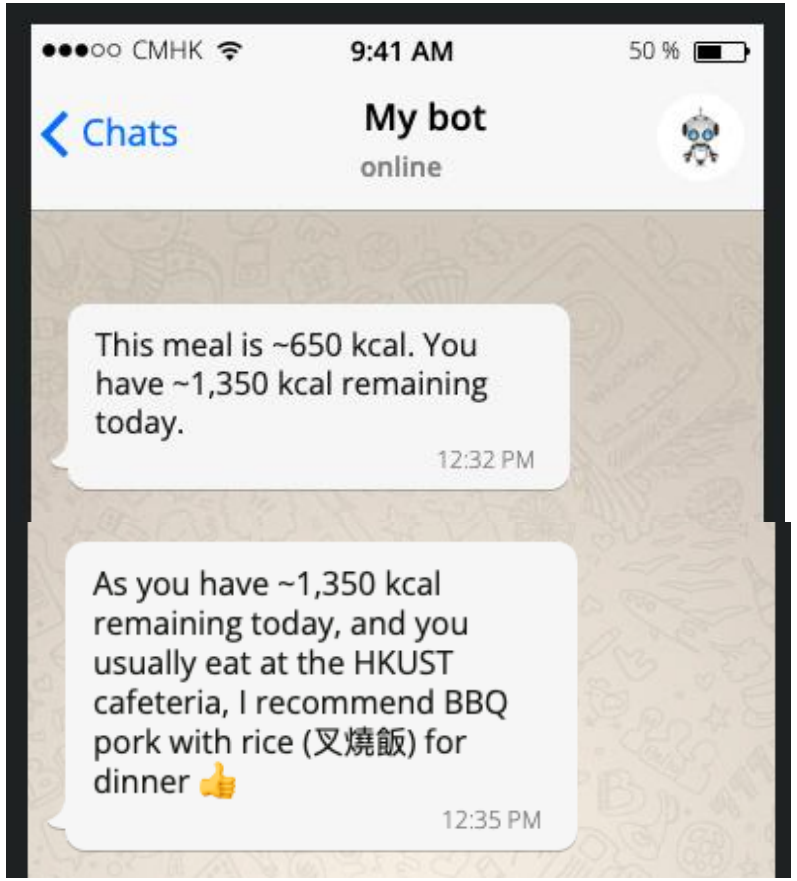
5 Secrets Y
ISN'T TELL

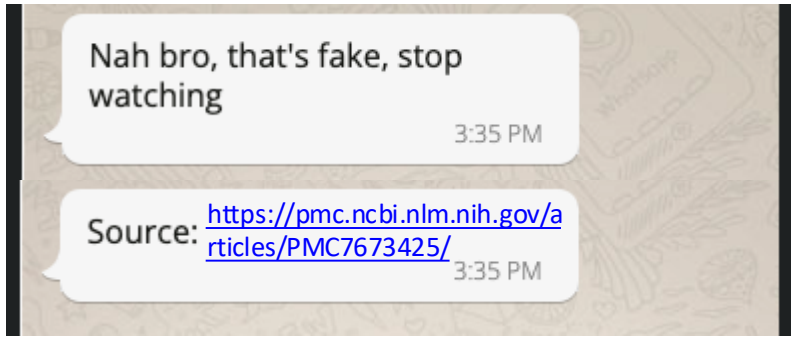
Overall Pick
NY Threads Professional White Lab Coat for Men, Long Sleeve Medical Coat
4.5 (6K)
1K+ bought in past month
$14.99 Typical: $17.99
Join Prime to get FREE delivery Tomorrow, Mar 25
Or Non-members get FREE delivery Sun, Mar 29 on $35 of items shipped by Amazon
Safer chemicals
Add to cart

各科医生绝对不会
告诉你的秘密
各科医生绝对不会告诉你的秘密！ #医生日常
#科普 #翘二郎腿
武汉协和皮肤科李延
2024-09-13
17

WHAT IF

?!
5 Secrets Your Doctor
ISN'T TELLING YOU

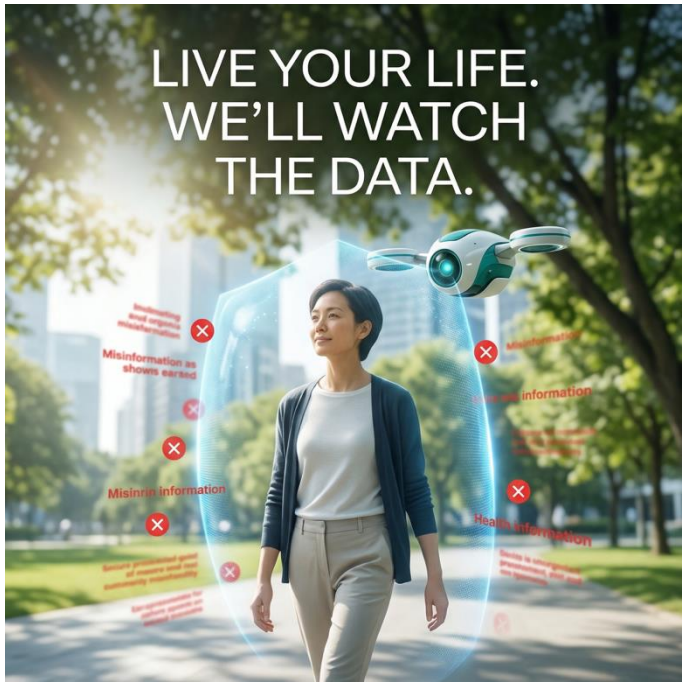

CMHK
9:41 AM
50 %
Chats
My bot
online
This meal is ~650 kcal. You have ~1,350 kcal remaining today.
12:32 PM
As you have ~1,350 kcal remaining today, and you usually eat at the HKUST cafeteria, I recommend BBQ pork with rice (叉燒飯) for dinner 👍
12:35 PM


Nah bro, that's fake, stop watching
3:35 PM
Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC7673425/
3:35 PM

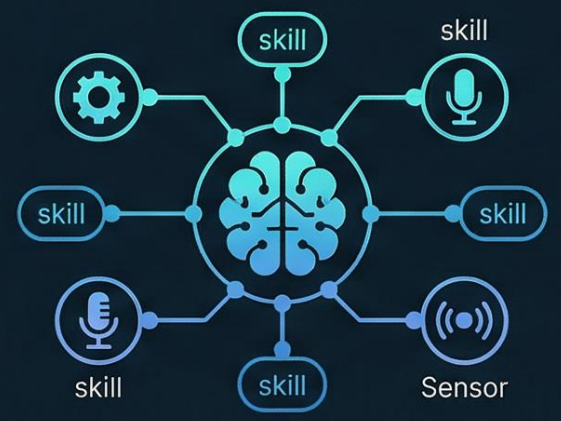
PRODUCT

Product Intro

Health Sentinel is an all-in-one AI Agent designed to watch over you when you aren't watching. It doesn't just record data; it thinks for you. By combining real-time environment sensing with advanced medical fact-checking, it acts as a digital filter, keeping you safe from both physical health risks and digital misinformation.



OPENCLAW AGENT FUNCTIONAL FRAMEWORK



Skill-Based Intelligence

Connects Agent with specialized 'Skills' for smart devices and environmental sensors.

The Truth Engine

Uses real-time medical databases to verify health claims and flag 'fake news'.

Personalized Memory

Remembers your unique medical history and allergies for tailored alerts.

How this Product resolve the needs

Filters Misinformation:

It cross-references online health claims with medical databases to shield users from "Information Explosion" and digital fake news.

Provides Personalized Safety:

It utilizes "Long-term Memory" of a user's medical history and allergies to provide uniquely tailored, safe advice.

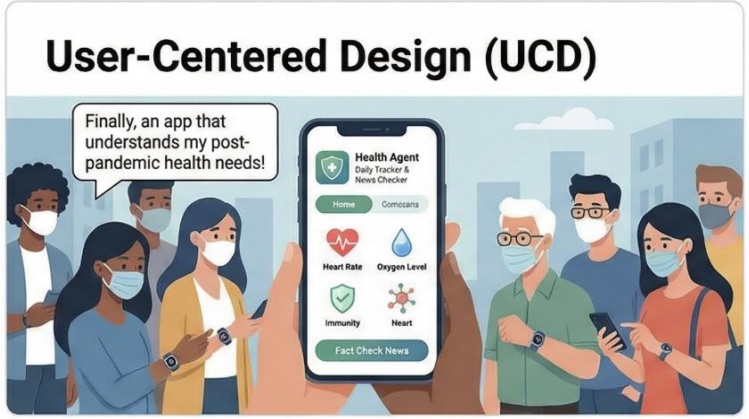
Reduces Cognitive Load:

By automating data monitoring through the OpenClaw framework, it handles the heavy "thinking," letting users focus on living their lives.

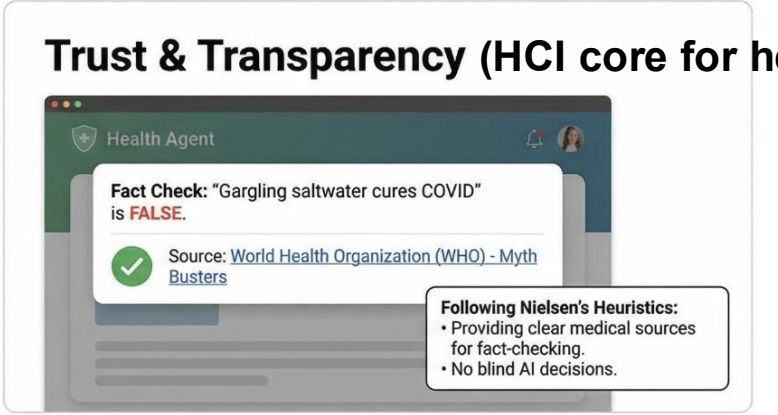
Ensures Full-Spectrum Vigilance:

It bridges environmental sensors (like air quality) with physiological data (like heart rate) for continuous, real-world protection.

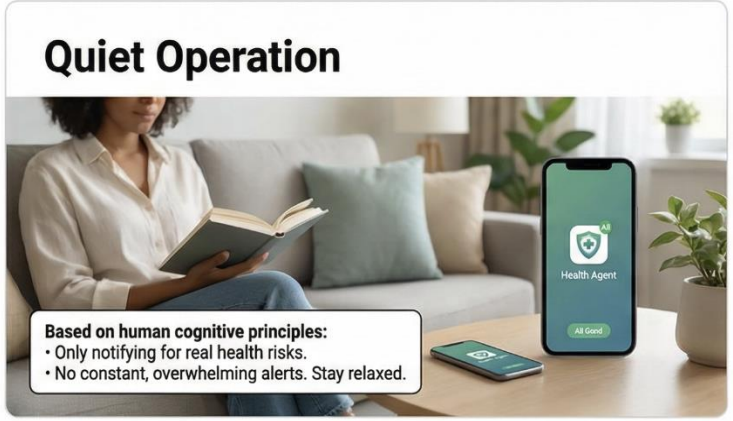
How this Product relates to HCI



- It’ s not tech-driven. We design the product for post-COVID user needs: daily health tracking and fighting fake health news, putting user pain points first.



- Following Nielsen’s heuristics, when the agent fact-checks news, it shows authoritative medical sources, which is critical for building trust instead of making blind AI decisions.

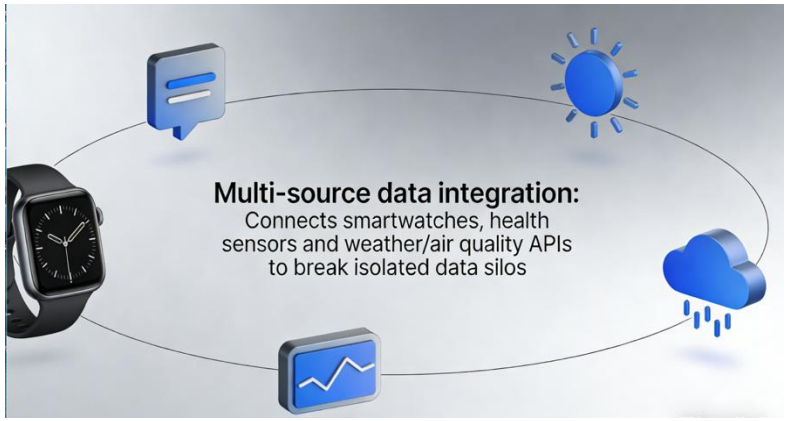


- Based on human cognitive principles, we use quiet operation: we only notify you for real health risks, avoiding information overload and alert fatigue.

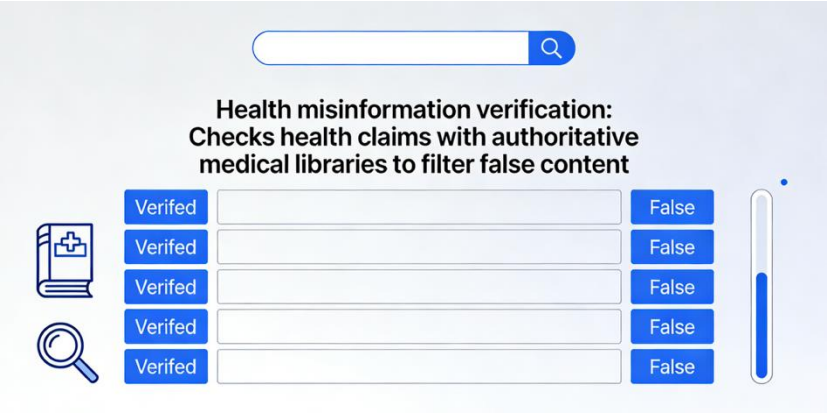
- Health data is highly sensitive. Therefore, we embed privacy-by-design: we enforce strict data protection and secure long-term memory, to meet user’s privacy needs.

Core Functions & UI Design

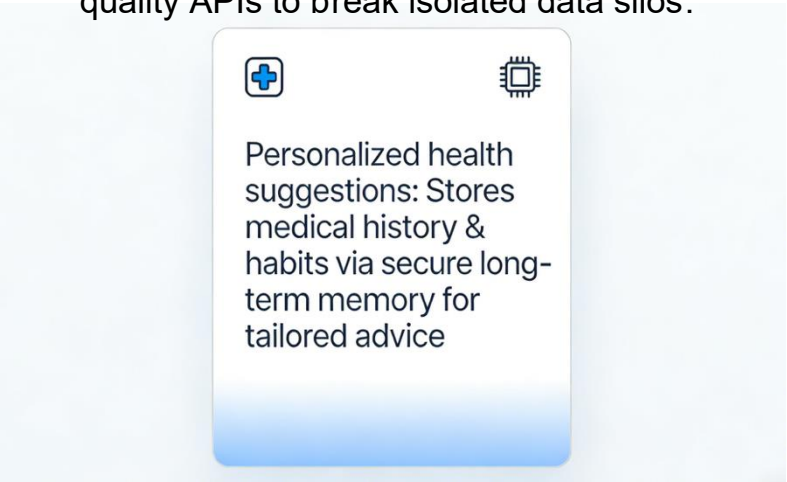
Core Functions



- **Multi-source data integration**: Connects smartwatches, health sensors and weather/air quality APIs to break isolated data silos.



- **Health misinformation verification**: Checks health claims with authoritative medical libraries to filter false content.

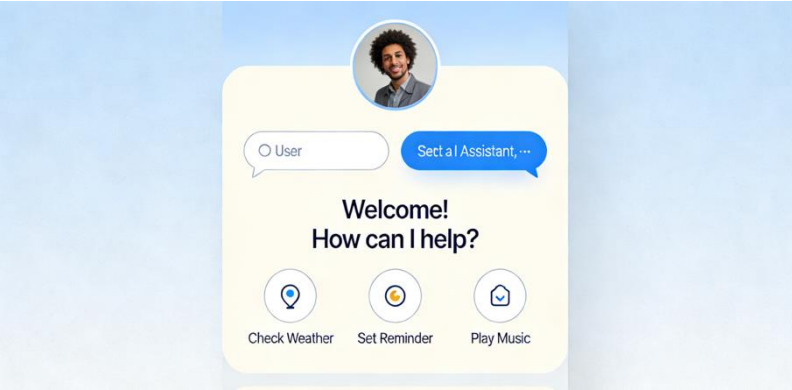


- **Personalized health suggestions:** Stores medical history & habits via secure long-term memory for tailored advice.

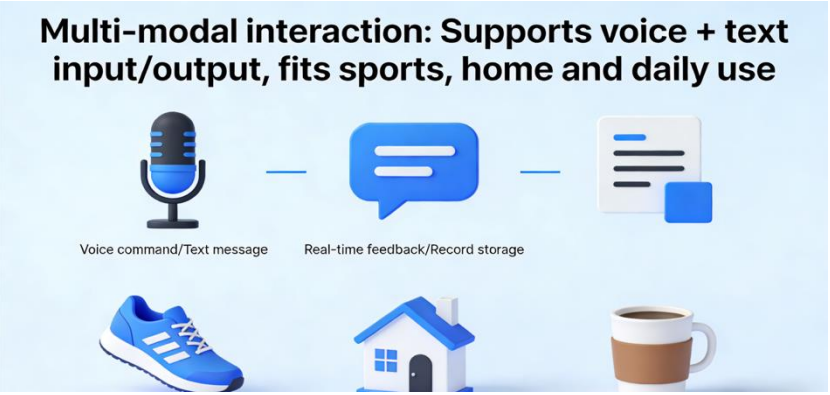
- **Smart risk alert:** Only notifies users of urgent health risks, avoiding unnecessary interruptions.

Core Functions & UI Design

UI Design

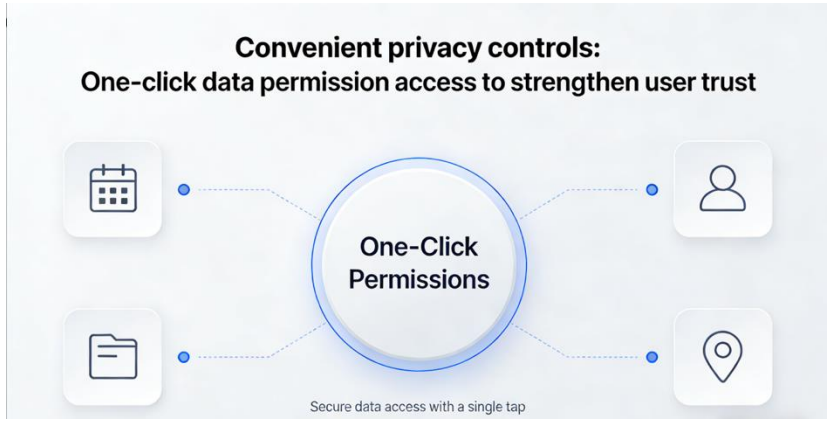


- **Simple & friendly interface**: Uses conversational assistant style, no complex operations for all users.



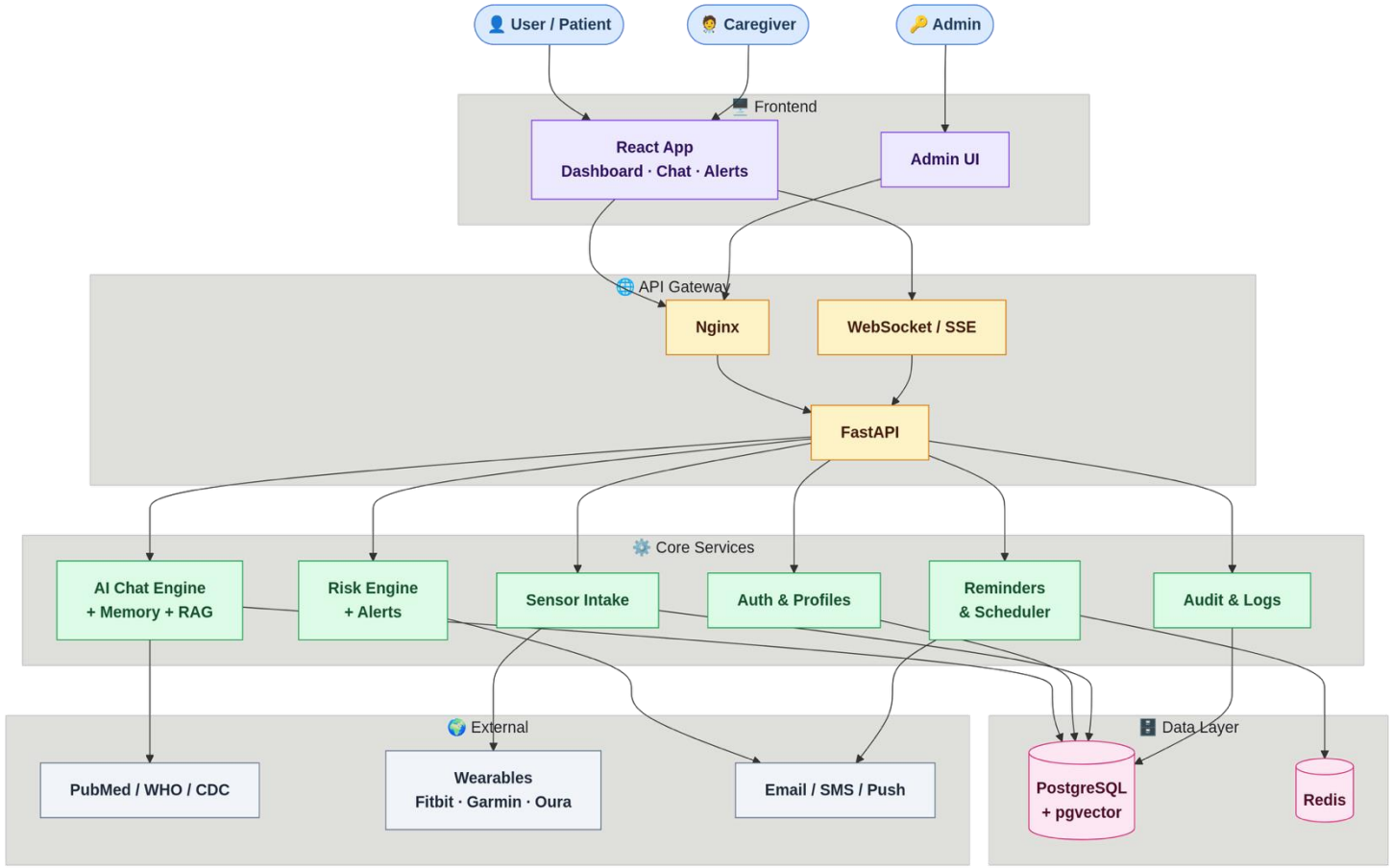
- **Multi-modal interaction**: Supports voice + text input/output, fits sports, home and daily use.

- **Transparent display**: Visualizes health data (heart rate, air quality) and shows clear fact-check sources.



- **Convenient privacy controls**: One-click data permission access to strengthen user trust.

Implementation design



Thank You

Q & A